

Project description Data Science Lab Winter Term 2025/26

- **Title:** please provide a short title (not more than one line)

Energy Consumption Patterns and Demand Forecasting in Slovak Households

- **Partner:** name of the partner company, Logo and Web page link that we can put on our Web page to name you as a collaboration partner in the data science lab.¹

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- **Data Coach:** Name and contact details for the data coach, email

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- **Project description:** general introduction to the topic domain and of the provided data, initial ideas for tasks that could be solved on the data in some bullet points.

Energy consumption patterns are increasingly relevant for energy providers, policymakers, and sustainability research. With the transition toward smarter energy systems and renewable integration, analyzing household-level consumption can provide valuable insights into demand response, tariff optimization, and regional differences in usage. In this project, students will work with a dataset of average 15-minute household energy consumption values (both active and reactive power) for 1000 anonymized households across the Slovak Republic in 2016. The dataset provides a rich basis for temporal analysis, clustering of household profiles and predictive modeling.

Tasks:

Exploratory Data Analysis (EDA):

Analyze daily and seasonal energy consumption patterns.
Identify households with unusual or extreme consumption behavior.
Explore differences between low-tariff and high-tariff periods.

Data Engineering & Enrichment:

Construct derived features (e.g., daily load profiles, weekday vs weekend patterns etc).
Aggregate or cluster households by ZIP code to study regional consumption.
Integrate additional datasets such as weather data, socio-economic indicators, or holidays.

Predictive Modeling:

Develop models to forecast short-term household energy demand (15-min, hourly, or daily).
Compare predictive approaches (time series models, regression)
Cluster households into consumption profiles (e.g., night owls, daytime users, high consumers).

Insights and Recommendations:

Interpret consumption drivers and feature importance.
Provide recommendations for energy efficiency measures, demand-response strategies, or tariff design.
Visualize findings through load curves, seasonal patterns, and household clusters (interactive dashboards or maps).

¹ Note that the project description is not published on the Web page publicly, but only accessible to the students of the course, we only name partner companies and their logos/links.

- **Dataset description:** Volume, format, which data already exists? Which additional data should the students find to combine with this existing data, if any? How can students access the data? Starting Points?

<https://data.mendeley.com/datasets/pns69yxgrp/2>

Students need to find additional metadata to combine with dataset (Metadata to weather, Socio-economic/demographic data etc))

- **Restrictions:** (e.g.) License restrictions, is there an NDA needed that the students need to sign? Will students need to sign any contract to work on the project and what are the conditions, in case?

No NDA / no contract, open data

- **Hardware:** Will you provide hardware or servers, how can the students connect to these? are the students supposed to work on your servers/premises or on their own laptops?

Own laptops

- **Software:** Which software/tools shall be used and are the needed tools openly available? If not, please provide information on whether/how the students can obtain (temporary) licenses from you.

R/Python (to be chosen by students)